Bent Marshak Waves - Cylindrical 2D Model

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1 Physical Setup and Geometry

We consider radiative heat transport in an optically thick medium inside a cylindrical tube. The longitudinal coordinate is denoted by z, and the transverse coordinate by the radial coordinate r. We assume full axisymmetry:

$$\frac{\partial}{\partial\theta} = 0.$$

The quantity of interest is

$$U(z, r, t) \equiv T^4(z, r, t),$$

which satisfies a diffusion–type equation under the assumptions of **constant opacity** and negligible hydrodynamic motion.

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2 Governing Equation

In the steady diffusion region behind the Marshak front, the radiation energy density satisfies Laplace's equation. In axisymmetric cylindrical coordinates this reads:

$$\frac{\partial^2 U}{\partial z^2} + \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial U}{\partial r}\right) = 0. \quad (1)$$

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3 Separation of Variables

We seek separable solutions of the form

$$U(z, r) = Z(z)R(r).$$

Substituting into Eq. (1) and dividing by ZR yields

$$\frac{1}{Z}\frac{d^2 Z}{dz^2} + \frac{1}{Rr}\frac{d}{dr}\left(r\frac{dR}{dr}\right) = 0.$$

Setting both terms equal to a separation constant $-\kappa^2$ gives

$$\frac{d^2Z}{dz^2} - \kappa^2 Z = 0, \tag{2}$$

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dR}{dr}\right) + \kappa^2 R = 0. \tag{3}$$

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4 Radial Eigenfunctions

The radial equation is Bessel's equation of order zero. Its regular solution at $r = 0$ is

$$R(r) = J_0(\kappa r).$$

The general axisymmetric solution therefore takes the form

$$U(z,r) = \sum_{n\geq 0} J_0(\kappa_n r)\left[A_n(t)e^{\kappa_n z} + B_n(t)e^{-\kappa_n z}\right]. \tag{4}$$

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5 Boundary Conditions

5.1 Axis Regularity

Regularity at the symmetry axis requires

$$\left.\frac{\partial U}{\partial r}\right|_{r=0} = 0.$$

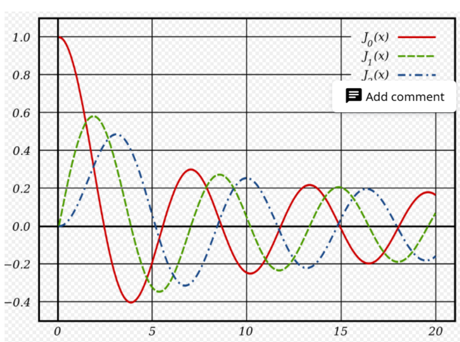

Figure 1: The first three Bessel functions of the first kind, J_0, J_1, and J_2.

For that reason, we want the r-spatial part to be at its peak at r=0, and not 0, which is only satisfied by $J_0(\kappa r)$.

5.2 Radiative Wall Losses

At the cylindrical wall $r = R$, radiative energy is lost according to a Marshak-type Robin boundary condition:

$$\left.\frac{\partial U}{\partial r}\right|_{r=R} = \frac{3}{4}\rho\kappa_{opacity}(a-1)\,U(R), \tag{5}$$

where a is the wall albedo.

Using the Bessel function identity (Eq. 46 and Eq. 45)

$$\frac{d}{dr}J_0(\kappa r) = -\kappa J_1(\kappa r),$$

the boundary condition yields the eigenvalue equation

$$\boxed{\kappa_n R J_1(\kappa_n R) = \varepsilon J_0(\kappa_n R),} \tag{6}$$

definition of small parameter:

$$\varepsilon \equiv \frac{3}{4}(1-a)\tau$$

where $\tau = \rho\kappa_{opacity}R$ is the optical depth.

Solutions are intersections of: $\kappa R J_1(\kappa R) = \varepsilon J_0(\kappa R)$. with small corrections.

This equation determines the discrete spectrum $\{\kappa_n\}$.

Small-loss limit. For $\varepsilon \ll 1$, we can express the the zeroth and first bessel functions using their series representation:

$$J_0(\kappa x) \approx 1 - \frac{(\kappa x)^2}{4},\;\; J_1(\kappa x) \approx \frac{(\kappa x)}{2} - \frac{(\kappa x)^3}{16}.$$

Substituting into Eq. (6) we get:

$$\kappa R\left(\frac{\kappa R}{2} - \frac{(\kappa R)^3}{16}\right) = \varepsilon\left(1 - \frac{(\kappa R)^2}{4}\right).$$

or leading order in κR,

$$\frac{(\kappa R)^2}{2} \approx \varepsilon, \implies \kappa_0 R \approx \sqrt{2\varepsilon}.$$

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6 Marshak Front Representation

The Marshak front position is expanded in the same radial eigenfunctions:

$$z_F(r,t) = \sum_{n\geq 0} c_n(t)\,J_0(\kappa_n r), \tag{7}$$

where $\{\kappa_n\}$ are determined by the Robin wall condition $\kappa R J_1(\kappa R) = \varepsilon J_0(\kappa R)$.

At leading order, only the lowest mode is retained:

$$z_F(r,t) \approx c_0(t)\,J_0(\kappa_0 r). \tag{8}$$

Front Evolution Equation: explicit leading-mode reduction

We now derive the ODE for $c_0(t)$ from the front energy balance. Following the Hurricane–Hammer steep-front closure, the front speed is proportional to the longitudinal radiative flux at the front:

$$\rho e\, \dot{z}_F = -\frac{4\sigma}{3\kappa_R \rho}\, \frac{\partial U}{\partial z}\bigg|_{z=z_F}. \tag{9}$$

Here κ_R is the (Rosseland) opacity and σ is the Stefan–Boltzmann constant; e is an effective energy-per-volume parameter for heating the material.

Step 1: leading eigenmode form of $U(z,r,t)$. From the separated solution (Eq. (4)) we retain only the lowest radial eigenfunction:

$$U(z,r,t) \approx J_0(\kappa_0 r)\left(A(t)e^{\kappa_0 z} + B(t)e^{-\kappa_0 z}\right). \tag{10}$$

We impose two longitudinal conditions (as in the planar derivation):

6.1 Source boundary at $z = 0$

From (4) at $z = 0$:

$$T_s^4 = \sum_{n=0}^{\infty} \left(A_n + B_n\right) J_0(k_n r). \tag{11}$$

Using orthogonality of $J_0(k_n r)$ on $[0, R]$ with weight r, this implies that to leading order only the fundamental mode carries the constant function:

$$A_0 + B_0 = T_s^4 + O(\epsilon), \qquad A_{n\geq 1} + B_{n\geq 1} = O(\epsilon). \tag{12}$$

(Exactly as in the planar derivation: the constant-in-r boundary drives primarily the lowest mode.)

6.2 Front boundary condition: weak-curvature implementation (full derivation)

The bent Marshak front is the moving surface

$$z = z_F(r,t),$$

at which the temperature drops sharply. In the steep-front approximation we impose

$$U\big(z_F(r,t), r, t\big) \approx 0.$$

Because the wall losses are assumed weak (high albedo), the curvature of the front is small. We therefore expand the front position about a mean (or center-line) position $c_0(t)$:

$$z_F(r,t) = c_0(t) + \delta z_F(r,t), \qquad \delta z_F = O(\epsilon). \tag{13}$$

Taylor expansion of the front boundary condition. Expand $U(z,r,t)$ about $z = c_0(t)$:

$$U\big(z_F(r,t),r,t\big) = U\big(c_0 + \delta z_F, r, t\big)$$
$$= U(c_0,r,t) + \delta z_F(r,t)\,\partial_z U(c_0,r,t) + \frac{1}{2}\delta z_F(r,t)^2\,\partial_z^2 U(c_0,r,t) + \cdots . \tag{14}$$

Imposing $U(z_F,r,t) \approx 0$ and retaining only the leading term in the weak-curvature limit gives

$$\boxed{U(c_0,r,t) = 0 \qquad \text{(leading order in curvature).}} \tag{15}$$

Higher-order corrections can be incorporated by keeping the $\delta z_F\,\partial_z U$ term; here we focus on the leading-order closure.

modal form of the solution in the diffusion region. In the quasi-steady diffusion region behind the front, U satisfies Laplace's equation and admits the separated expansion

$$U(z,r,t) = \sum_{n=0}^{\infty} J_0(k_n r)\Big(A_n(t)e^{k_n z} + B_n(t)e^{-k_n z}\Big). \tag{16}$$

impose the leading-order front condition mode-by-mode. Insert $z = c_0$ into (16) and enforce (15):

$$0 = U(c_0,r,t) = \sum_{n=0}^{\infty} J_0(k_n r)\Big(A_n e^{k_n c_0} + B_n e^{-k_n c_0}\Big).$$

Since the set $\{J_0(k_n r)\}$ forms an orthogonal basis on $0 \le r \le R$ (with weight r), the coefficient of each mode must vanish:

$$\boxed{A_n(t)e^{k_n c_0(t)} + B_n(t)e^{-k_n c_0(t)} = 0 \qquad \forall n.} \tag{17}$$

This immediately gives

$$B_n(t) = -A_n(t)e^{2k_n c_0(t)}. \tag{18}$$

rewrite the longitudinal dependence in hyperbolic-sine form. Start with the generic z-dependence of mode n:

$$A_n e^{k_n z} + B_n e^{-k_n z}.$$

Substitute (18):

$$\begin{aligned} A_n e^{k_n z} + B_n e^{-k_n z} &= A_n e^{k_n z} - A_n e^{2k_n c_0} e^{-k_n z} \\ &= A_n \left(e^{k_n z} - e^{k_n(2c_0 - z)}\right) \\ &= A_n e^{k_n c_0}\left(e^{k_n(z-c_0)} - e^{-k_n(z-c_0)}\right) \\ &= 2A_n e^{k_n c_0}\sinh\big(k_n(z-c_0)\big). \end{aligned} \tag{19}$$

It is convenient to factor out a normalization $\sinh(k_n c_0)$ so that the remaining coefficient is dimensionless and (later) directly tied to the source boundary condition. Define a new amplitude $\widetilde{A}_n(t)$ by

$$\widetilde{A}_n(t) \equiv 2A_n(t)e^{k_n c_0(t)} \sinh\big(k_n c_0(t)\big). \tag{20}$$

Then (19) becomes the compact "sinh-form"

$$\boxed{A_n e^{k_n z} + B_n e^{-k_n z} = \widetilde{A}_n(t) \frac{\sinh\big(k_n(z - c_0(t))\big)}{\sinh\big(k_n c_0(t)\big)}.} \tag{21}$$

This expression automatically satisfies the front condition because at $z = c_0$ the numerator vanishes.

insert back into the full field expansion. Substituting (21) into (16) yields

$$U(z, r, t) = \sum_{n=0}^{\infty} J_0(k_n r)\, \widetilde{A}_n(t) \frac{\sinh\big(k_n(z - c_0(t))\big)}{\sinh\big(k_n c_0(t)\big)}. \tag{22}$$

normalize by the source temperature and define dimensionless amplitudes. Let T_s be the imposed source temperature (for constant drive) so that the natural scale is T_s^4. Define dimensionless coefficients

$$\alpha_n(t) \equiv -\frac{\widetilde{A}_n(t)}{T_s^4}. \tag{23}$$

Then (22) becomes

$$\boxed{\frac{U(z, r, t)}{T_s^4} = -\sum_{n=0}^{\infty} \alpha_n(t)\, J_0(k_n r) \frac{\sinh\big(k_n(z - c_0(t))\big)}{\sinh\big(k_n c_0(t)\big)}.} \tag{24}$$

This is exactly the cylindrical analogue of the paper's planar expression (their Eq. (27)), with $\cos(\cdot)$ replaced by $J_0(\cdot)$ and transverse eigenvalues set by the Robin condition.

how α_n are fixed by the source boundary condition. At $z = 0$ the boundary condition is $U(0, r, t) = T_s^4$ (constant in r for a uniform drive). Plugging $z = 0$ into (24) gives

$$1 = \frac{U(0, r, t)}{T_s^4} = -\sum_{n=0}^{\infty} \alpha_n\, J_0(k_n r) \frac{\sinh\big(-k_n c_0\big)}{\sinh\big(k_n c_0\big)} = \sum_{n=0}^{\infty} \alpha_n\, J_0(k_n r),$$

since $\sinh(-x) = -\sinh(x)$. Therefore

$$\boxed{1 = \sum_{n=0}^{\infty} \alpha_n\, J_0(k_n r) \qquad (0 \le r \le R),} \tag{25}$$

i.e. the constant function 1 is expanded in the $J_0(k_n r)$ basis. This uniquely determines the α_n via Bessel orthogonality (with weight r). In the high-albedo limit, the lowest mode is nearly uniform, so

$$\alpha_0 = 1, \qquad \alpha_{n\geq 1} = O(\epsilon),$$

matching the weak-loss ordering used in the perturbation theory. Therefore, to leading order we can set $\alpha_0 = 1$ and $\alpha_{n\geq 1} = 0$, giving the leading-mode approximation

$$\boxed{\frac{U(z,r,t)}{T_s^4} \approx -\alpha_0 J_0(k_0 r)\frac{\sinh(k_0(z-c_0(t)))}{\sinh(k_0 c_0(t))} = J_0(k_0 r)\frac{\sinh(k_0(c_0(t)-z))}{\sinh(k_0 c_0(t))}.} \tag{26}$$

Step 2: compute the front gradient $\partial_z U|_{z=z_F}$. Differentiate (26) with respect to z:

$$\partial_z U(z,r,t) \approx U_s(t)\, J_0(\kappa_0 r)\, \frac{\partial}{\partial z}\left[\frac{\sinh(\kappa_0(c_0-z))}{\sinh(\kappa_0 c_0)}\right] = \\ -U_s(t)\, J_0(\kappa_0 r)\, \frac{\kappa_0 \cosh(\kappa_0(c_0-z))}{\sinh(\kappa_0 c_0)}.$$

Evaluating at the front $z = c_0(t)$ gives

$$\partial_z U\big|_{z=c_0(t)} \approx -U_s(t)\, J_0(\kappa_0 r)\, \frac{\kappa_0}{\sinh\big(\kappa_0 c_0(t)\big)}. \tag{27}$$

Step 3: project the front balance onto the lowest eigenmode. Equation (9) is a *pointwise* statement along the front. In the leading-mode truncation we enforce it in a weighted (Galerkin) sense by multiplying both sides by $J_0(\kappa_0 r)$ and integrating over the cross section with cylindrical weight $r\,dr$:

$$\int_0^R \rho e\, \dot{z}_F(r,t)\, J_0(\kappa_0 r)\, r\, dr = -\frac{4\sigma}{3\kappa_R \rho}\int_0^R \partial_z U\big|_{z=z_F}\, J_0(\kappa_0 r)\, r\, dr. \tag{28}$$

Using $z_F(r,t) \approx c_0(t) J_0(\kappa_0 r)$, the left-hand side becomes

$$\int_0^R \rho e\, \dot{c}_0(t)\, J_0^2(\kappa_0 r)\, r\, dr = \rho e\, \dot{c}_0(t)\, \mathcal{N}_0, \qquad \mathcal{N}_0 \equiv \int_0^R J_0^2(\kappa_0 r)\, r\, dr.$$

Using (27), the right-hand side becomes

$$-\frac{4\sigma}{3\kappa_R \rho}\int_0^R \left(-U_s(t)\, J_0(\kappa_0 r)\frac{\kappa_0}{\sinh(\kappa_0 c_0)}\right) J_0(\kappa_0 r)\, r\, dr = \frac{4\sigma}{3\kappa_R \rho}\frac{U_s(t)\kappa_0}{\sinh(\kappa_0 c_0)}\, \mathcal{N}_0.$$

Canceling the common factor $\mathcal{N}_0$ yields the closed ODE

$$e\, \dot{c}_0(t) = \frac{4\sigma}{3\kappa_R \rho^2}\, \frac{U_s(t)\kappa_0}{\sinh\big(\kappa_0 c_0(t)\big)}. \tag{29}$$

Step 4: define an effective diffusion constant D_M. For a constant drive $U_s(t) = U_0$ (or for slow variation treated quasi-statically), we define the effective constant

$$D_M \equiv \frac{8\sigma U_0}{3e\,\kappa_R \rho^2}, \tag{30}$$

so that (29) becomes

$$\dot{c}_0(t) = \frac{D_M \kappa_0}{2} \frac{1}{\sinh\big(\kappa_0 c_0(t)\big)}. \tag{31}$$

Solution for the Isotropic Mode: integration to $\cosh^{-1}$

Multiply (31) by $\kappa_0 \sinh(\kappa_0 c_0)$:

$$\kappa_0 \sinh(\kappa_0 c_0)\,\dot{c}_0 = \frac{D_M \kappa_0^2}{2}.$$

Recognizing the left-hand side as a time derivative,

$$\frac{d}{dt}\cosh(\kappa_0 c_0) = \kappa_0 \sinh(\kappa_0 c_0)\,\dot{c}_0,$$

we obtain

$$\frac{d}{dt}\cosh(\kappa_0 c_0) = \frac{D_M \kappa_0^2}{2}. \tag{32}$$

Integrating from $t = 0$ with $c_0(0) = 0$ gives

$$\cosh(\kappa_0 c_0(t)) = 1 + \frac{D_M \kappa_0^2 t}{2}.$$

Therefore,

$$\boxed{c_0(t) = \frac{1}{\kappa_0}\cosh^{-1}\left(1 + \frac{D_M \kappa_0^2\, t}{2}\right).} \tag{33}$$

Finally, the leading-order front shape is

$$\boxed{z_F(r,t) \approx c_0(t)\, J_0(\kappa_0 r) = \frac{1}{\kappa_0}\cosh^{-1}\left(1 + \frac{D_M \kappa_0^2\, t}{2}\right) J_0(\kappa_0 r).} \tag{34}$$

One can make the derivation even better if defining $D = D_M(1 + \frac{\varepsilon}{2})$ (higher order accuracy):

$$\boxed{z_F(r,t) \approx c_0(t)\, J_0(\kappa_0 r) = \frac{1}{\kappa_0}\cosh^{-1}\left(1 + \frac{D \kappa_0^2\, t}{2}\right) J_0(\kappa_0 r).} \tag{35}$$

Time-dependent drive $U_s(t)$ (non-constant boundary temperature)

So far, the closed-form solution (33) was obtained assuming a *constant* drive $U_s(t) = U_0$ (equivalently, constant boundary temperature T_s so that $U_s = T_s^4$). We now generalize the derivation to an *arbitrary* time-dependent drive $U_s(t)$.

Where $U_s(t)$ enters. The only place where the drive appears in the leading-mode reduction is through the boundary condition at $z = 0$,

$$U(0, r, t) = U_s(t),$$

which implies $A(t) + B(t) = U_s(t)$ (Eq. (12)). Repeating the same algebra as in (27) (with $U_0 \to U_s(t)$), we obtain the front gradient

$$\partial_z U\big|_{z=c_0(t)} \approx -U_s(t)\, J_0(\kappa_0 r)\, \frac{\kappa_0}{\sinh\big(\kappa_0 c_0(t)\big)}. \tag{36}$$

Therefore, the Galerkin projection step (Eq. (28)) yields the same ODE structure as before, but with an explicit $U_s(t)$ factor:

$$\dot{c}_0(t) = \frac{\kappa_0}{\sinh\big(\kappa_0 c_0(t)\big)}\, \frac{4\sigma}{3e\,\kappa_R \rho^2}\, U_s(t). \tag{37}$$

Define a time-dependent diffusion coefficient. It is convenient to define

$$D_M(t) \equiv \frac{8\sigma}{3e\,\kappa_R \rho^2}\, U_s(t), \tag{38}$$

so that Eq. (37) becomes

$$\dot{c}_0(t) = \frac{D_M(t)\,\kappa_0}{2}\, \frac{1}{\sinh\big(\kappa_0 c_0(t)\big)}. \tag{39}$$

Integration for general $U_s(t)$. Multiply Eq. (39) by $\kappa_0 \sinh(\kappa_0 c_0)$ to obtain

$$\frac{d}{dt} \cosh(\kappa_0 c_0(t)) = \frac{\kappa_0^2}{2}\, D_M(t).$$

Integrating from 0 to t with $c_0(0) = 0$ gives

$$\boxed{\cosh\big(\kappa_0 c_0(t)\big) = 1 + \frac{\kappa_0^2}{2} \int_0^t D_M(t')\, dt' = 1 + \frac{4\sigma\,\kappa_0^2}{3e\,\kappa_R \rho^2} \int_0^t U_s(t')\, dt'.} \tag{40}$$

Hence,

$$\boxed{c_0(t) = \frac{1}{\kappa_0} \cosh^{-1}\!\left(1 + \frac{\kappa_0^2}{2} \int_0^t D_M(t')\, dt'\right) = \frac{1}{\kappa_0} \cosh^{-1}\!\left(1 + \frac{4\sigma\,\kappa_0^2}{3e\,\kappa_R \rho^2} \int_0^t U_s(t')\, dt'\right).} \tag{41}$$

Finally, the leading-order cylindrical front shape becomes

$$\boxed{z_F(r, t) \approx c_0(t)\, J_0(\kappa_0 r) = \frac{J_0(\kappa_0 r)}{\kappa_0} \cosh^{-1}\!\left(1 + \frac{4\sigma\,\kappa_0^2}{3e\,\kappa_R \rho^2} \int_0^t U_s(t')\, dt'\right).} \tag{42}$$

Special case: power-law drive. If $U_s(t) = U_0 \left(\frac{t}{t_0}\right)^p$ for $t \geq 0$, then

$$\int_0^t U_s(t')\,dt' = \frac{U_0}{p+1}\,\frac{t^{p+1}}{t_0^p},$$

and Eq. (41) becomes

$$c_0(t) = \frac{1}{\kappa_0}\cosh^{-1}\left(1 + \frac{4\sigma\,\kappa_0^2}{3e\,\kappa_R\rho^2}\cdot\frac{U_0}{p+1}\,\frac{t^{p+1}}{t_0^p}\right).$$

This reduces to the constant-drive expression when $p = 0$.

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7 Physical Interpretation

- The transverse structure is governed by $J_0(\kappa_0 r)$ rather than $\cos(k_0 y)$.
- Radiative wall losses introduce curvature and retard the front.
- Cylindrical geometry enhances edge cooling relative to planar geometry.
- The leading-order dynamics are entirely controlled by the lowest eigenvalue κ_0.

The correspondence between planar and cylindrical formulations is:

$$\cos(ky) \;\longleftrightarrow\; J_0(\kappa r), \qquad \tan(kL) \;\longleftrightarrow\; \frac{\kappa R J_1(\kappa R)}{J_0(\kappa R)}.$$

All higher-order perturbation theory carries over with this substitution.

8 Usful bessel function identities

Series Representation:

$$J_n(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\,\Gamma(m+n+1)}\left(\frac{x}{2}\right)^{2m+n} \tag{43}$$

Proparties:

$$2nJ_n(x) = x(J_{n-1}(x) + J_{n+1}(x)) \tag{44}$$

$$J_n(-x) = (-1)^n J_n(x), \quad J_{-n} = (-1)^n J_n(x) \tag{45}$$

Differentiation,

$$\frac{d}{dx}J_n(x) = \frac{1}{2}(J_{n-1}(x) - J_{n+1}(x)) \tag{46}$$

$$\frac{d}{dx}\left[x^n J_n(x)\right] = x^n J_{n-1}(x), \qquad \frac{d}{dx}\left[x^{-n} J_n(x)\right] = -x^{-n} J_{n+1}(x) \tag{47}$$

Asymptotic form for large and small x:

$$J_n(x) \approx \frac{1}{n!}\left(\frac{x}{2}\right)^n, \quad x \ll 1 \tag{48}$$

$$J_n(x) \approx \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{n\pi}{2} - \frac{\pi}{4}\right), \quad x \gg 1 \tag{49}$$

And lastly the orthogonality relation:

$$\int_0^R r J_n\left(\frac{\alpha_m r}{R}\right) J_n\left(\frac{\alpha_k r}{R}\right) dr = \frac{R^2}{2}\left[J_{n+1}(\alpha_m)\right]^2 \delta_{mk} \tag{50}$$

where α_m is the m'th root of $J_n(x)$.

Albedo Derivation

Consider a gold cylindrical shell surrounding foam.

Let

$$F_{\text{in}} = \sigma_{SB} T_s^4 \tag{51}$$

$$F_{\text{out}} = \sigma_{SB} T_{\text{obs}}^4 \tag{52}$$

The detector measures the re-emitted flux:

$$\sigma_{SB} T_{\text{obs}}^4 = \sigma_{SB} T_s^4 + \frac{F(0,t)}{2} \tag{53}$$

Thus,

$$\sigma_{SB} T_s^4 - \frac{F(0,t)}{2} = \sigma_{SB} T_{\text{obs}}^4 \tag{54}$$

Since

$$F(0,t) = \dot{E}_w \tag{55}$$

we define the albedo:

$$a(t) = \frac{\text{reflected flux}}{\text{incident flux}} = \frac{F_{\text{in}}}{F_{\text{out}}} = \frac{\sigma_{SB} T_0^4 + \frac{\dot{E}_w}{2}}{\sigma_{SB} T_0^4 - \frac{\dot{E}_w}{2}} \tag{56}$$